

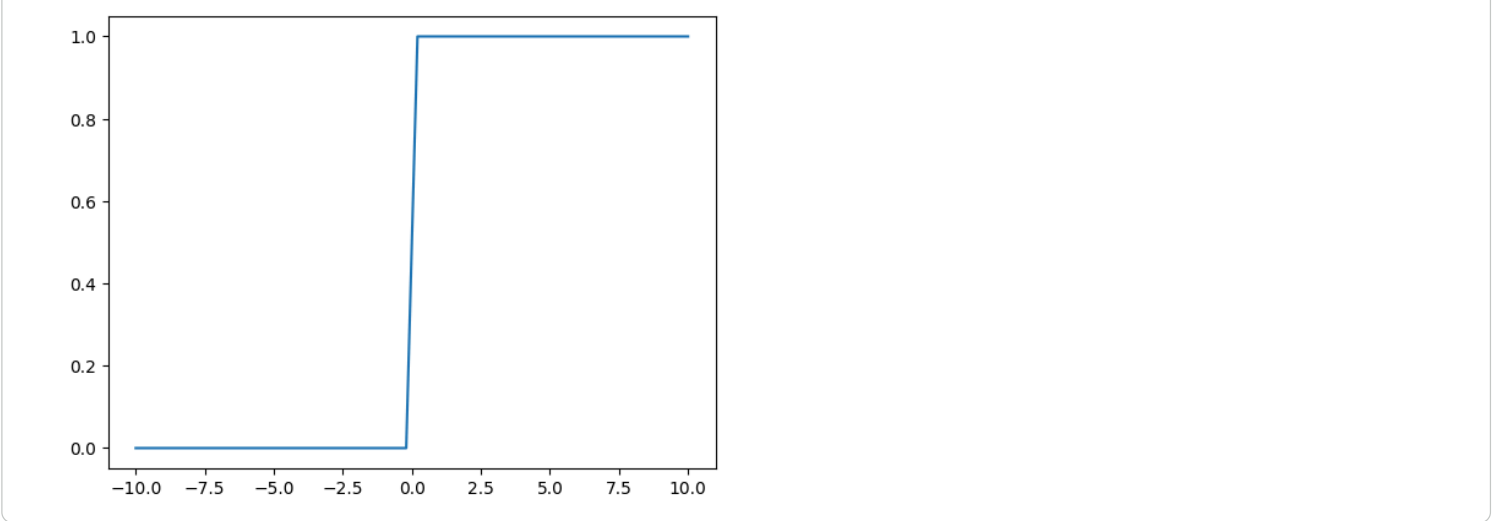
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import math
```

```
x = np.linspace(-10, 10, 50)
```

```
def step(x):  
    return 1 if x>=0 else 0
```

```
binary = []
for i in x:
    y = step(i)
    binary.append(y)
binary
```

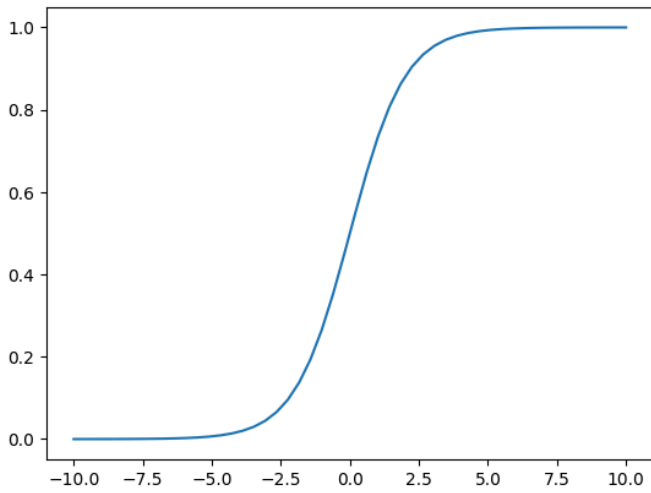
```
y = [step(i) for i in x]
plt.plot(x, y)
plt.show()
```



Sigmoid Function

```
def sigmoid(x):  
    y = 1/(1 + math.exp(-x))  
    return y
```

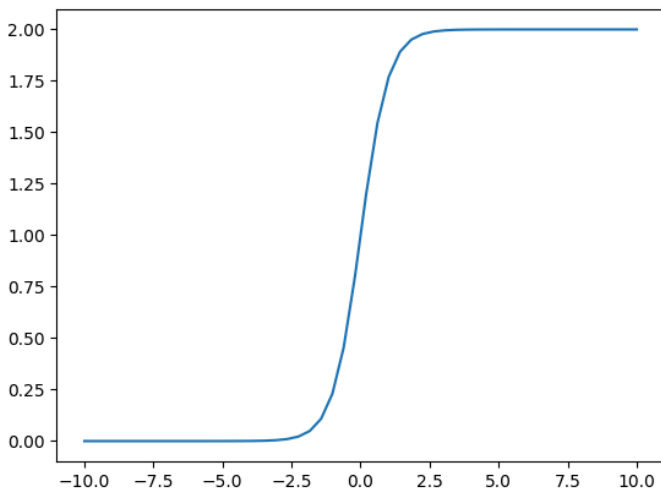
```
y = [sigmoid(i) for i in x]  
plt.plot(x, y)  
plt.show()
```



✓ Tanh Function

```
def tanh(x):  
    return (2 / (1 + math.exp(-2*x)))
```

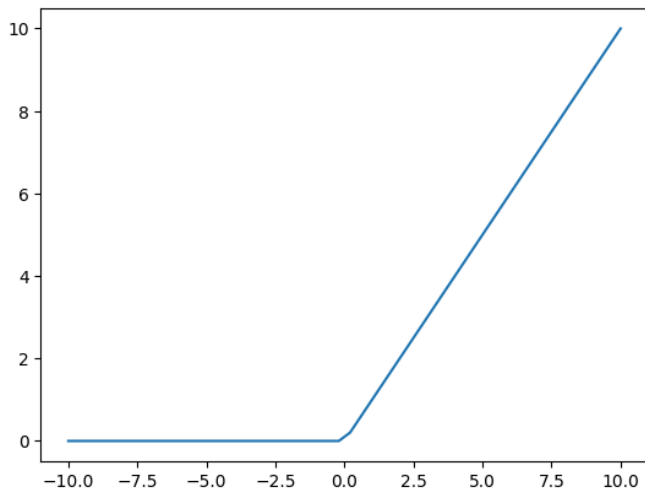
```
y = [tanh(i) for i in x]  
plt.plot(x, y)  
plt.show()
```



✓ ReLU Function

```
def relu(x):  
    return x if x >= 0 else 0 # max(x, 0)
```

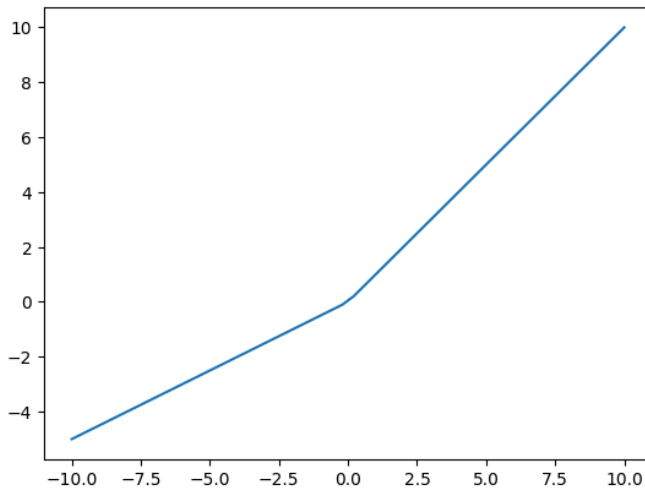
```
y = [relu(i) for i in x]  
plt.plot(x, y)  
plt.show()
```



Leaky Relu Function

```
def leaky_relu(x, alpha=0.5):
    return x if x >= 0 else alpha*x # max(x, alpha*x)
```

```
y = [leaky_relu(i) for i in x]
plt.plot(x, y)
plt.show()
```



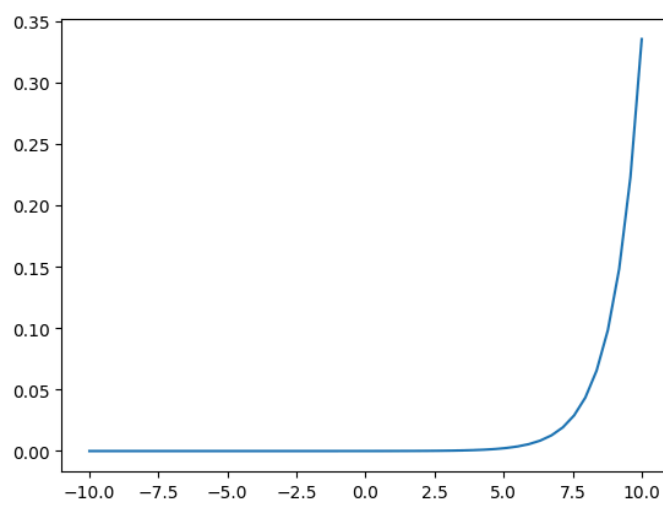
```
binary = []
for i in x:
    y = leaky_relu(i)
    binary.append(y)
binary
```

[Show hidden output](#)

SoftMax Function

```
def softmax(x): # there is an error
    e_x_sum = 0
    for i in x:
        e_x_sum += math.exp(i)
    return [math.exp(i)/e_x_sum for i in x]
```

```
y = softmax(x)
plt.plot(x, y)
plt.show()
```



Softplus

```
def softplus(x):  
    return np.log(1 + np.exp(x))
```

```
y = [softplus(i) for i in x]  
plt.plot(x, y)  
plt.show()
```

